

Predicting the Severity of Political Violence in Kenya

A Machine Learning Approach Using Historical Conflict Fatality Data

ACLED + Kenya Census | 1997–2026

47 Counties • 16,549 Observations

Moringa School | Data Science Programme

PROJECT SNAPSHOT

DOMAIN

Conflict Studies / Political Science / Machine Learning

DATE RANGE

January 1997 – June 2026

DATASET SIZE

16,549 county-week observations

MODELS

Linear Regression, Random Forest, XGBoost, Gradient Boosting

DATA SOURCE

ACLED + Kenya Census Population & Density Data

GEOGRAPHIC SCOPE

47 Counties of Kenya

TARGET VARIABLE

FATALITIES (count per county-week)

METRICS

MAE, RMSE, R² Score

THE PROBLEM

Kenya has experienced recurring political violence since 1991. Rich ACLED conflict data exists — yet little structured ML work has been applied to predict fatality severity at county level.



Quantify Risk

Identify conditions likely to produce high-fatality outcomes



Early Warning

Flag high-risk counties before violence escalates



Policy Guidance

Interpretable feature importance rankings for decision-makers

⚠️ Distributional Challenge

72.6% of observations are ZERO fatalities — Max = 281. Mean = 0.95. Predicting rare high-lethality events is analytically non-trivial and practically urgent.

OBJECTIVES

01

Identify statistically significant predictors of fatalities using feature importance scores and correlation analysis

02

Quantify the marginal effect of COUNTY_POPULATION and COUNTY_DENSITY on predicted fatality counts

03

Build & evaluate 4 supervised ML regression models — ranked by MAE, RMSE, and R^2 on a held-out test set

04

Produce a county-level vulnerability index ranking all 47 Kenyan counties by predicted high-fatality risk

05

Deliver interpretable SHAP-based feature importance mapped to actionable policy recommendations

DATA UNDERSTANDING

Data Sources

ACLED

Armed Conflict Location & Event Data
Kenya records 1997 – June 2026
Event type, sub-type, fatality counts

Kenya Census

County-level COUNTY_POPULATION
& COUNTY_DENSITY from KNBS
Demographic enrichment features

Key Distributional Facts

16,549

County-week observations

47

Kenyan counties covered

29 years

Temporal range (1997–2026)

72.6%

Zero-fatality rows

281

Maximum fatalities in one week

KEY DATASET FEATURES

Column	Type	Description
ADMIN1	Categorical	County name — geographic unit of analysis
EVENT_TYPE	Categorical	Type of conflict event (6 categories)
SUB_EVENT_TYPE	Categorical	Granular event sub-type (24 categories)
EVENTS	Numeric	Count of incidents per county-week
FATALITIES ★	Numeric TARGET	Deaths per county-week — model output
COUNTY_POPULATION	Numeric	Total county population (census)
COUNTY_DENSITY	Numeric	Population density (people/km ²)
WEEK_DATE	DateTime	Week of observation (1997–2026)
YEAR / MONTH	Derived	Temporal features extracted from WEEK_DATE
CENTROID_LAT/LON	Numeric	County centroid coordinates — spatial feature

DATA PREPARATION

Columns Dropped

- REGION — constant value 'Eastern Africa'
- COUNTRY — constant value 'Kenya'
- WEEK — duplicate Excel serial date
- DISORDER_TYPE — redundant grouping of EVENT_TYPE

Missing Value Treatment

- POPULATION_EXPOSURE: 29.4% missing
- Imputed using COUNTY_POPULATION as proxy
- Census population substituted where exposure unavailable

Feature Engineering

- Extract: month, quarter, week-of-year, is_election_year
- Fatality rate per 100k (EDA use only)
- Lag features: events_lag_1, fatalities_lag_1
- Label / Target Encoding for EVENT_TYPE & SUB_EVENT_TYPE
- Log-transform COUNTY_POPULATION & COUNTY_DENSITY

Train / Test Split

- 80/20 temporal split
- Training: 1997–2021
- Test: 2022–2026
- Simulates real-world deployment on unseen future data

MODELLING APPROACH

1

Linear Regression

Baseline

Interpretable baseline. Coefficients reveal predictor directions & magnitude.

2

Random Forest

Ensemble

Captures non-linear interactions. Native feature importance via mean decrease in impurity.

3

XGBoost

Boosting

Gradient-boosted trees with L1/L2 regularisation. Well-suited to sparse, skewed targets.

4

Gradient Boosting

Boosting

Sequential ensemble (sklearn). Direct comparison counterpart to XGBoost.



Hyperparameter Tuning: GridSearchCV / RandomizedSearchCV with 5-fold CV — tuning `n_estimators`, `max_depth`, `learning_rate`, `min_samples_split`

EVALUATION & INTERPRETABILITY

MAE

Mean Absolute Error

Interpretable in original units (fatalities). Robust to extreme outliers.

RMSE

Root Mean Squared Error

Penalises large prediction errors heavily. Critical given high-fatality tail events.

R^2

Coefficient of Determination

Proportion of fatality variance explained. Target: ≥ 0.60 on held-out test set.

Success Criteria

- At least one model achieves $R^2 \geq 0.60$ on held-out test set
- RMSE improves on Linear Regression baseline by $\geq 20\%$
- Feature importance outputs align with domain knowledge
- County-level vulnerability rankings validated against known hotspots

Interpretability Tools

SHAP Values

Global + individual feature importance for best model

Partial Dependence Plots

Marginal effect of COUNTY_DENSITY & POPULATION

Vulnerability Index

Ranked choropleth map of Kenya's 47 counties

DEPLOYMENT & TOOLS

GitHub Repository

All notebooks, datasets & scripts at
github.com/richardoketch32806-eng

Jupyter Notebook

End-to-end pipeline from raw ingestion through
evaluation

README

Non-technical findings summary, vulnerability
map & policy recommendations

Serialised Model

Best model saved as .pkl for inference on new
ACLED data exports

Vulnerability CSV

County-level risk scores exported for
humanitarian use

Tech Stack

Language: Python 3.11 / Jupyter Notebook

Preprocessing: scikit-learn · Label Encoding · StandardScaler

Visualisation: matplotlib · seaborn · folium / geopandas

Data: pandas · numpy · openpyxl

Modelling: sklearn · XGBoost · SHAP

Version Control: GitHub

EXPECTED IMPACT

OPERATIONAL

Pre-positioning of security and medical resources in high-risk counties ahead of expected violence events

POLICY

Evidence-based justification for targeted peace interventions — serving NCIC, County Security Committees, UNHCR

RESEARCH

Reproducible ML baseline for Kenya — extensible by other researchers to additional African conflict contexts